



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

November 27, 2025

Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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Frequentist View

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- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

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Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

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■ *Variance* or second raw moment: $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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- *Arbitrary Distributions:* obtained from uniformly distributed numbers by various methods

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$$I_{\nu}(T, \beta) = B_{\nu}(T) (1 - e^{-\tau_{\nu}(\beta)}) \approx B_{\nu}(T) \tau_{\nu}(\beta) = B_{\nu}(T) \Sigma \kappa_{\nu}(\beta) = \frac{2h\Sigma\kappa_0}{c^2\nu_0^{\beta}} \frac{\nu^{3+\beta}}{\exp\left(\frac{h\nu}{kT}\right) - 1}$$

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 - suppose we have no prior information on the parameter distribution

Example

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■ *Likelihood:*

$$L(I_v | v, \sigma_v, \theta) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_{v,i}^2}} \exp\left(-\frac{(I_{v,i} - I_v(\theta))^2}{2\sigma_{v,i}^2}\right)$$

Example

■ *Likelihood:*

$$\log L(I_v | v, \sigma_v, \theta) = -\frac{N}{2} \log(2\pi) - \sum_{i=1}^N \sigma_{v,i} - \frac{1}{2} \sum_{i=1}^N \frac{(I_{v,i} - I_v(\theta))^2}{\sigma_{v,i}^2}$$

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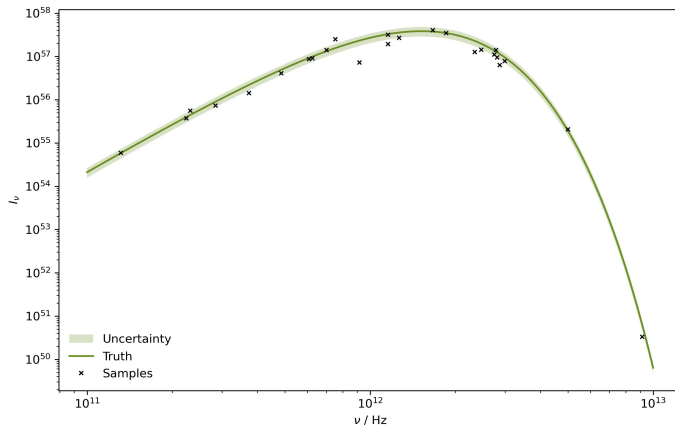
- *Posterior:*

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- *Objective:* minimise the $\chi^2(\theta)$ measure to maximise the posterior probability

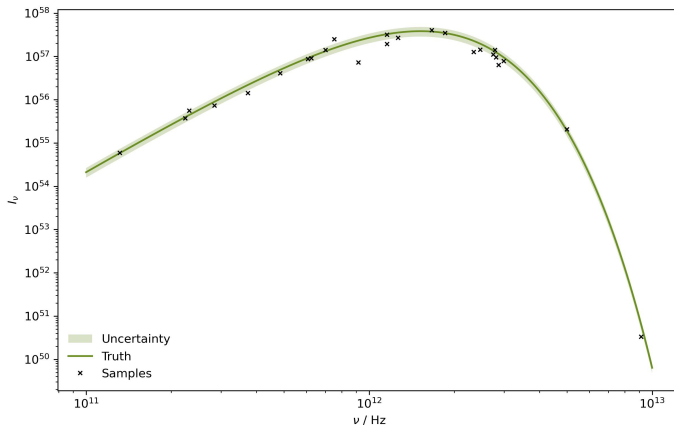
Measurements

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How to we proceed from this data to
maximise the likelihood for our
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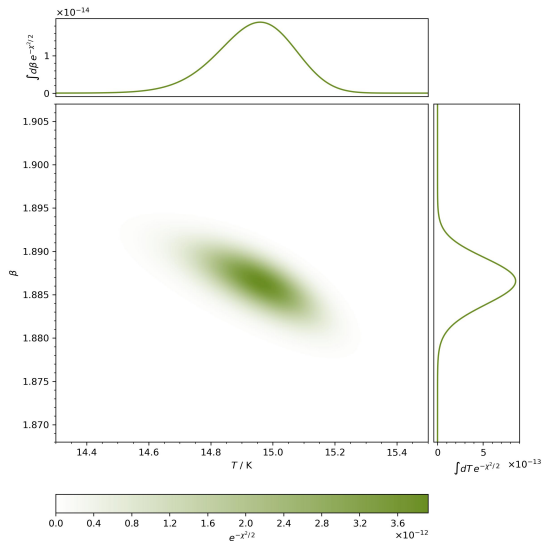
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- *Problem:* as D increases we have N^D gridpoints, and with more complex parameter distributions, this procedure quickly becomes intractable due to computational limits

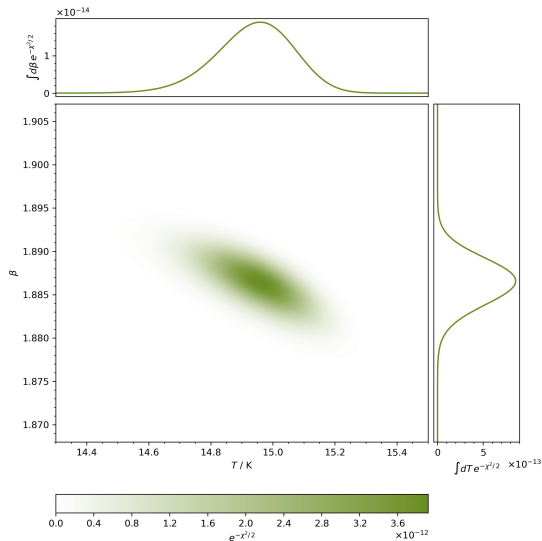
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Can we think of more elegant ways to move even through high dimensional parameter spaces?





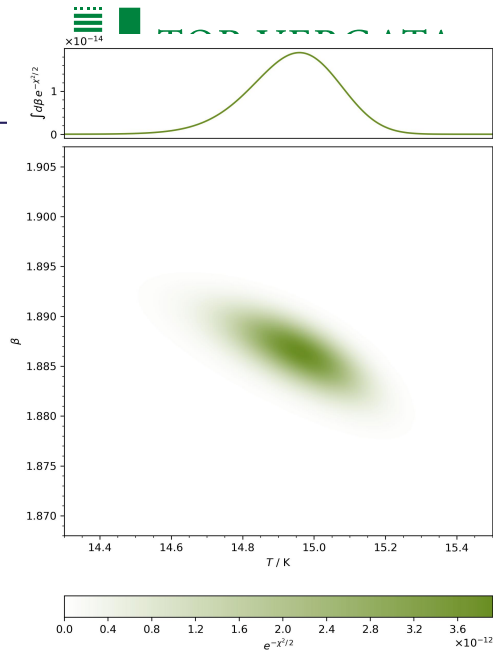
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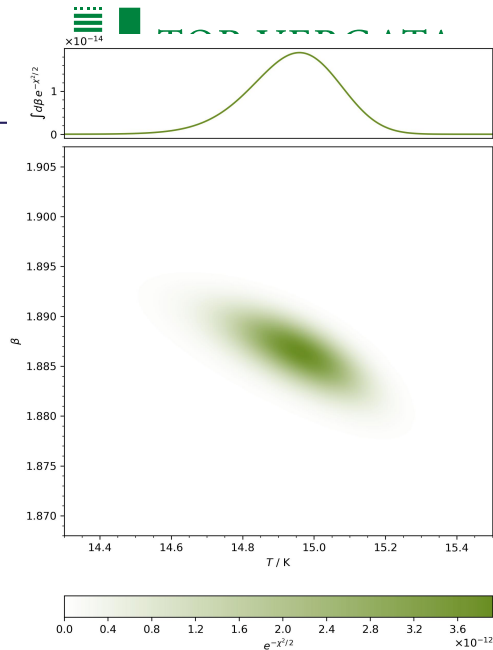
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Where I Studied

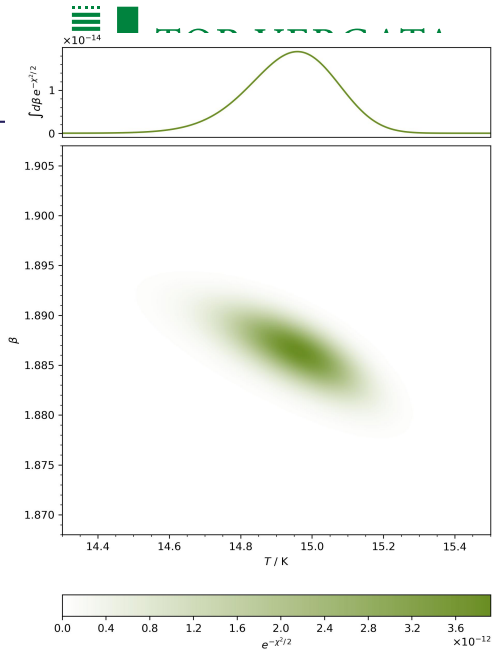
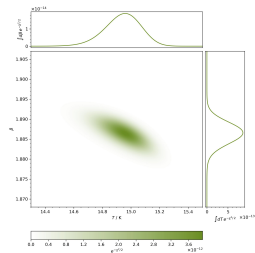
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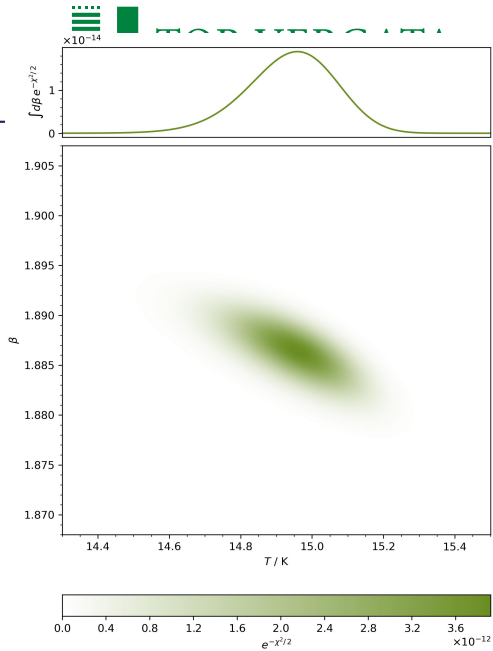
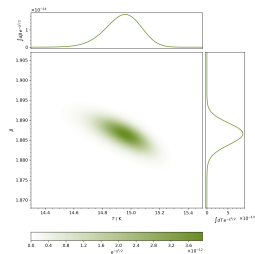
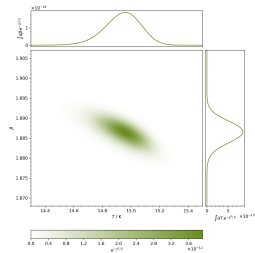
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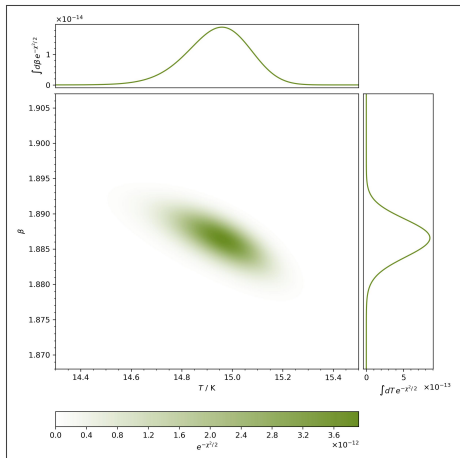
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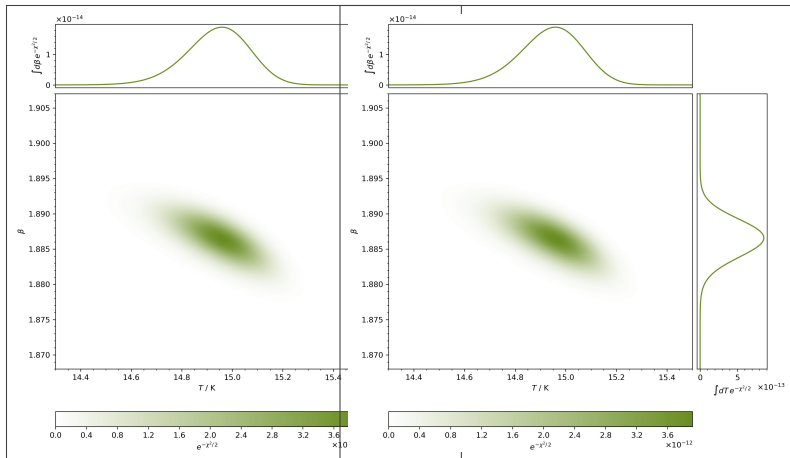
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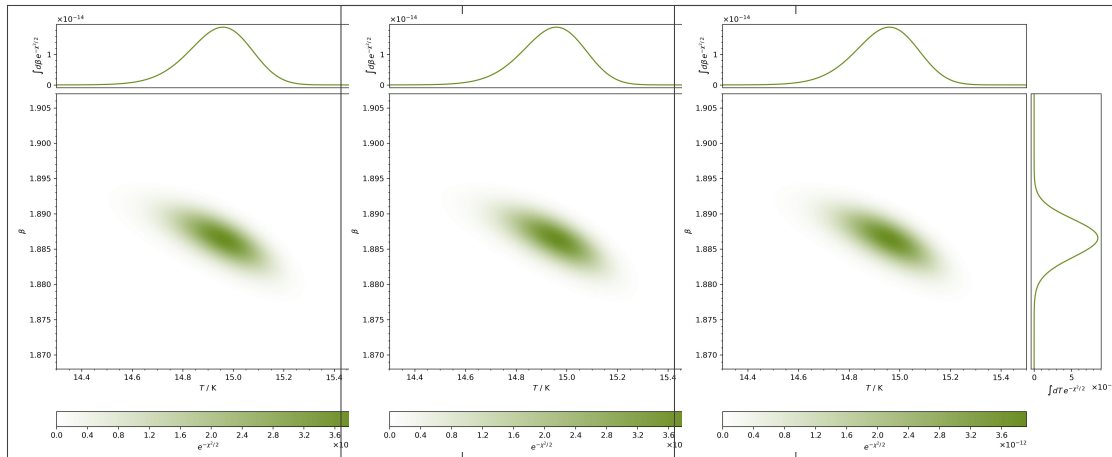
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Why Astrophysics

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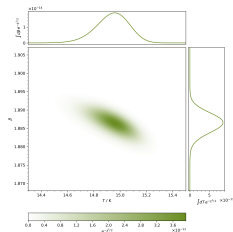
- Interesting

Why Astrophysics

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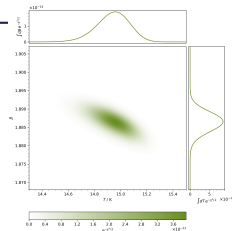
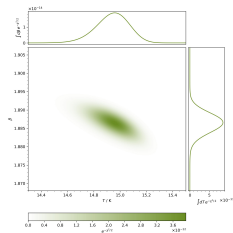
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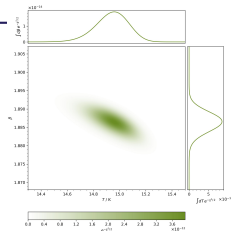
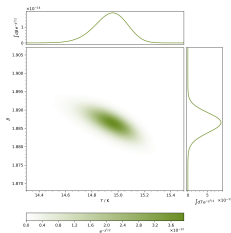
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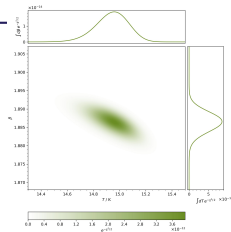
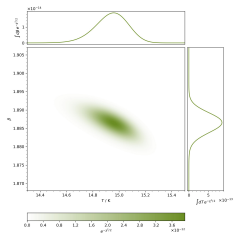
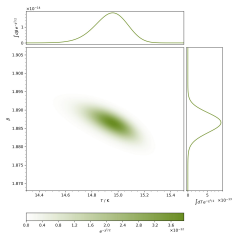
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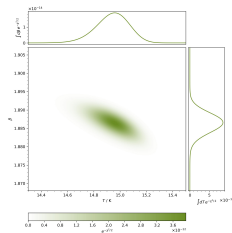
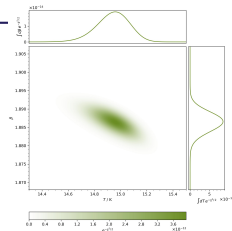
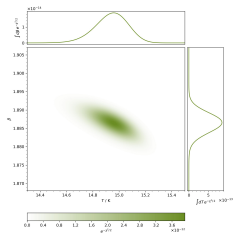
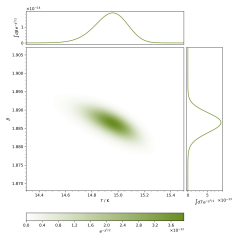
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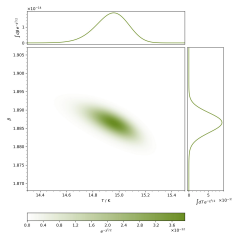
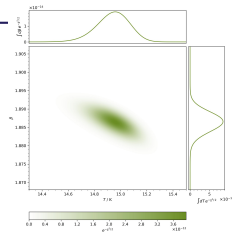
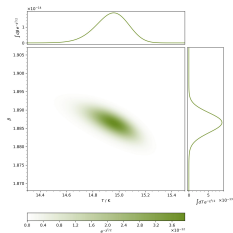
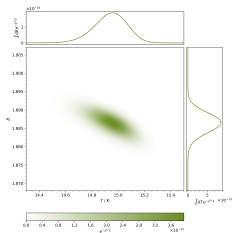
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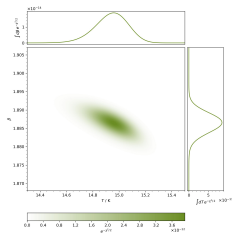
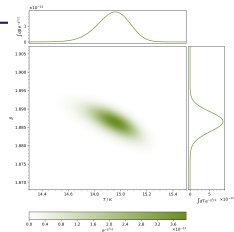
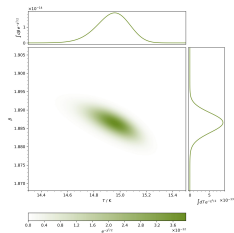
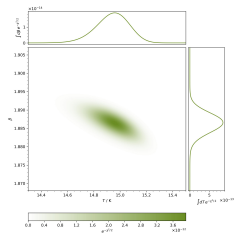
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 - Readily Observable
- Interdisciplinary



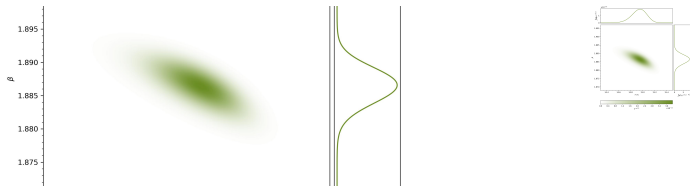
Why Astrophysics

- Interesting
 - Awesome Scales
 - Readily Observable
- Interdisciplinary
- International

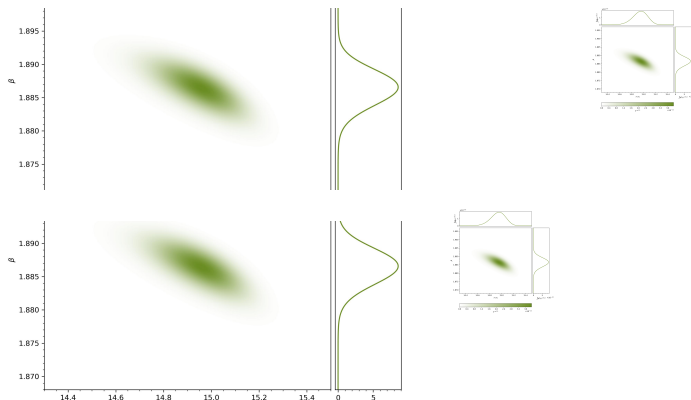


Where I Have Been

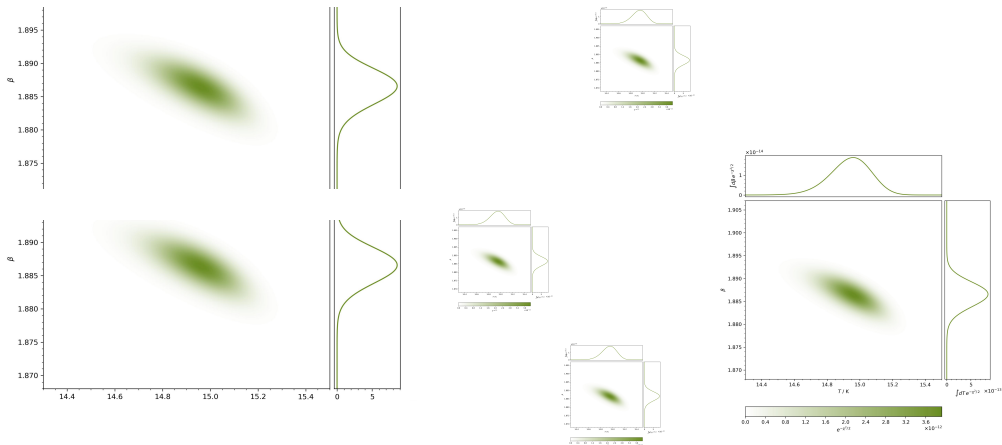
Where I Have Been



Where I Have Been



Where I Have Been



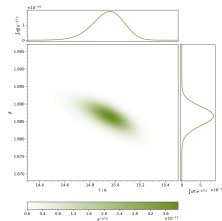
Other Interests

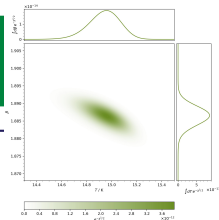
Other Interests

- Nature Photography

Other Interests

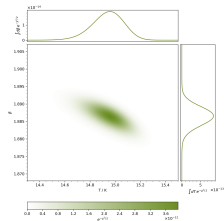
- Nature Photography





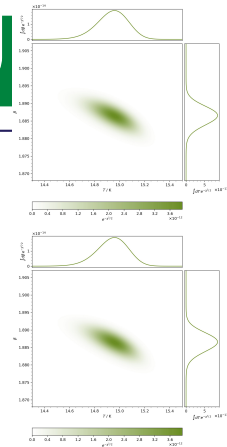
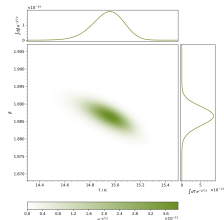
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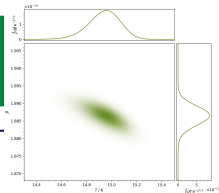
- Nature Photography



Other Interests

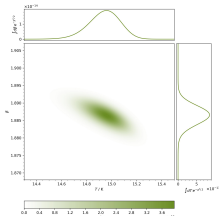
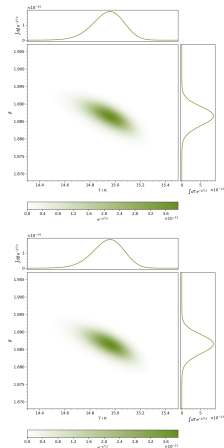
- Nature Photography

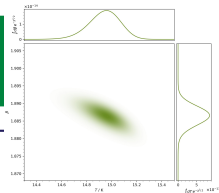




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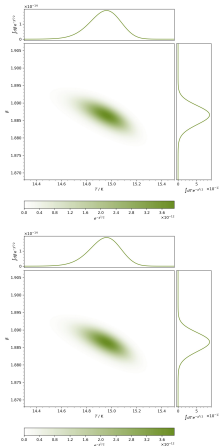
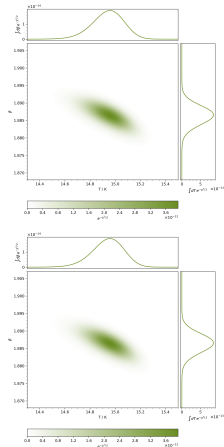
■ Nature Photography





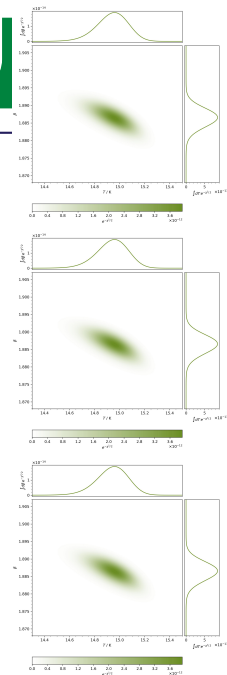
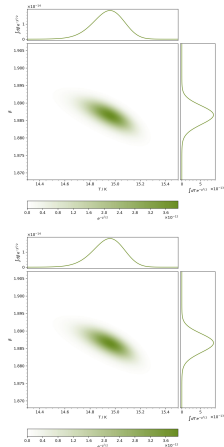
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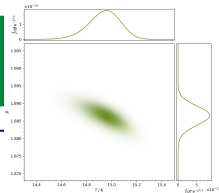
■ Nature Photography



Other Interests

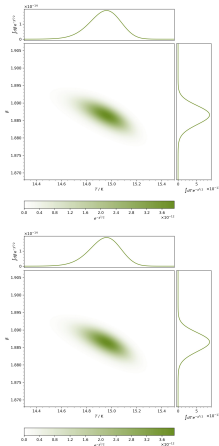
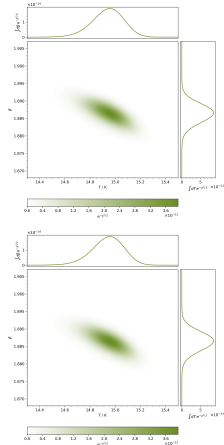
- Nature Photography
- Hiking & Cycling

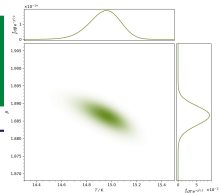




Other Interests

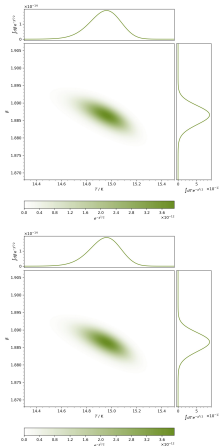
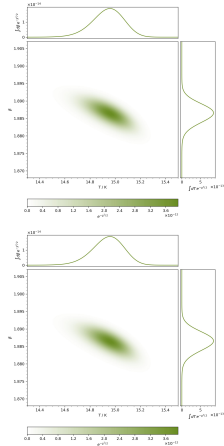
- Nature Photography
- Hiking & Cycling
- Programming

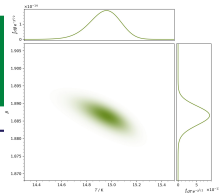




Other Interests

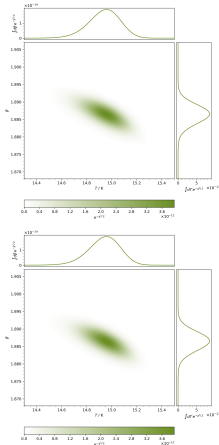
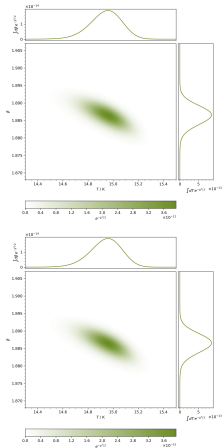
- Nature Photography
- Hiking & Cycling
- Programming
- Politics





Other Interests

- Nature Photography
- Hiking & Cycling
- Programming
- Politics
- Meeting New People



Thank You!